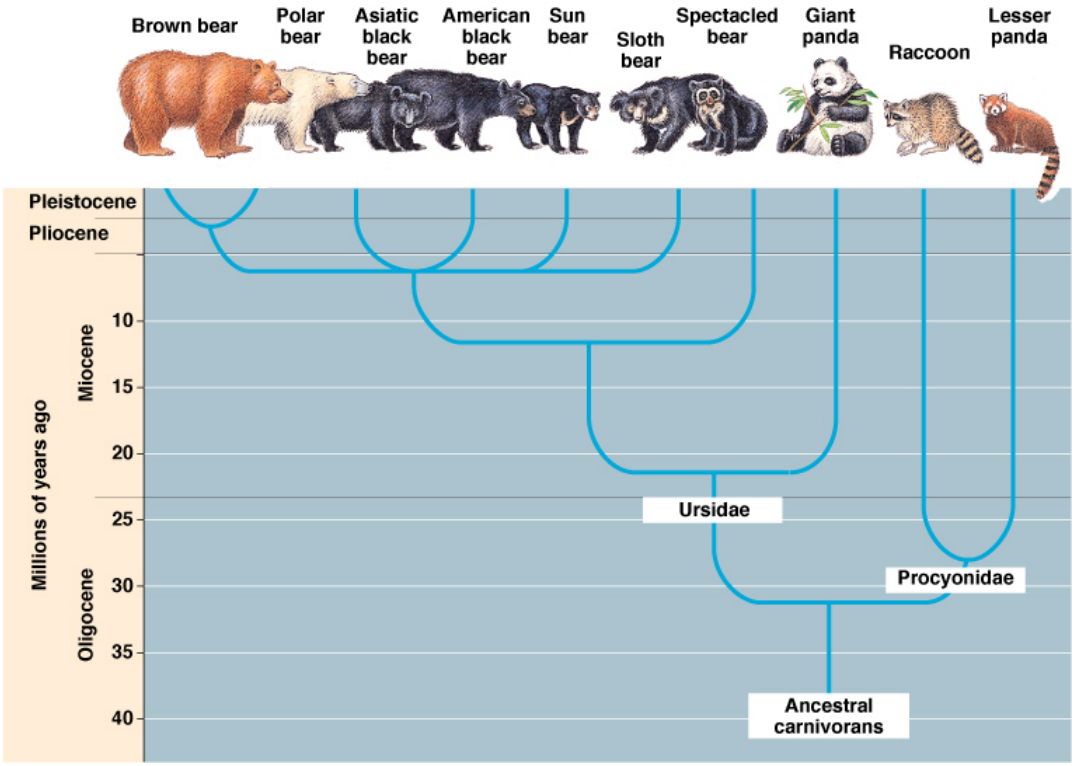
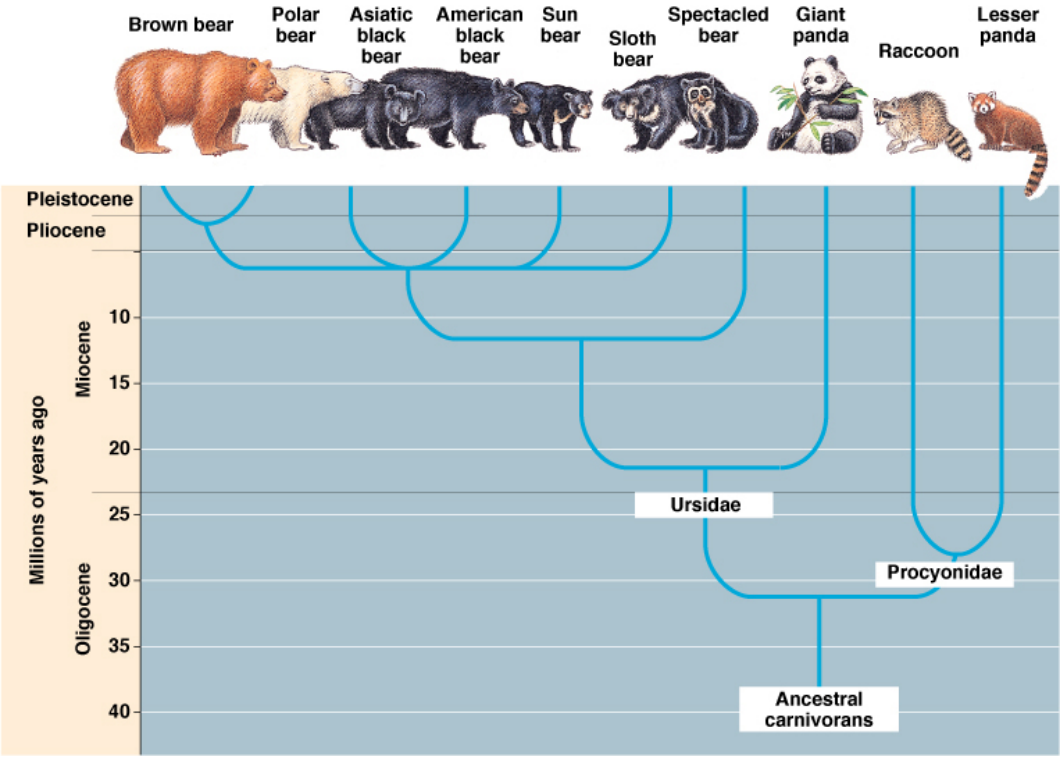
Lecture 13: *Phylogenetic Trees*



Trees of Life, UPGMA

Outline

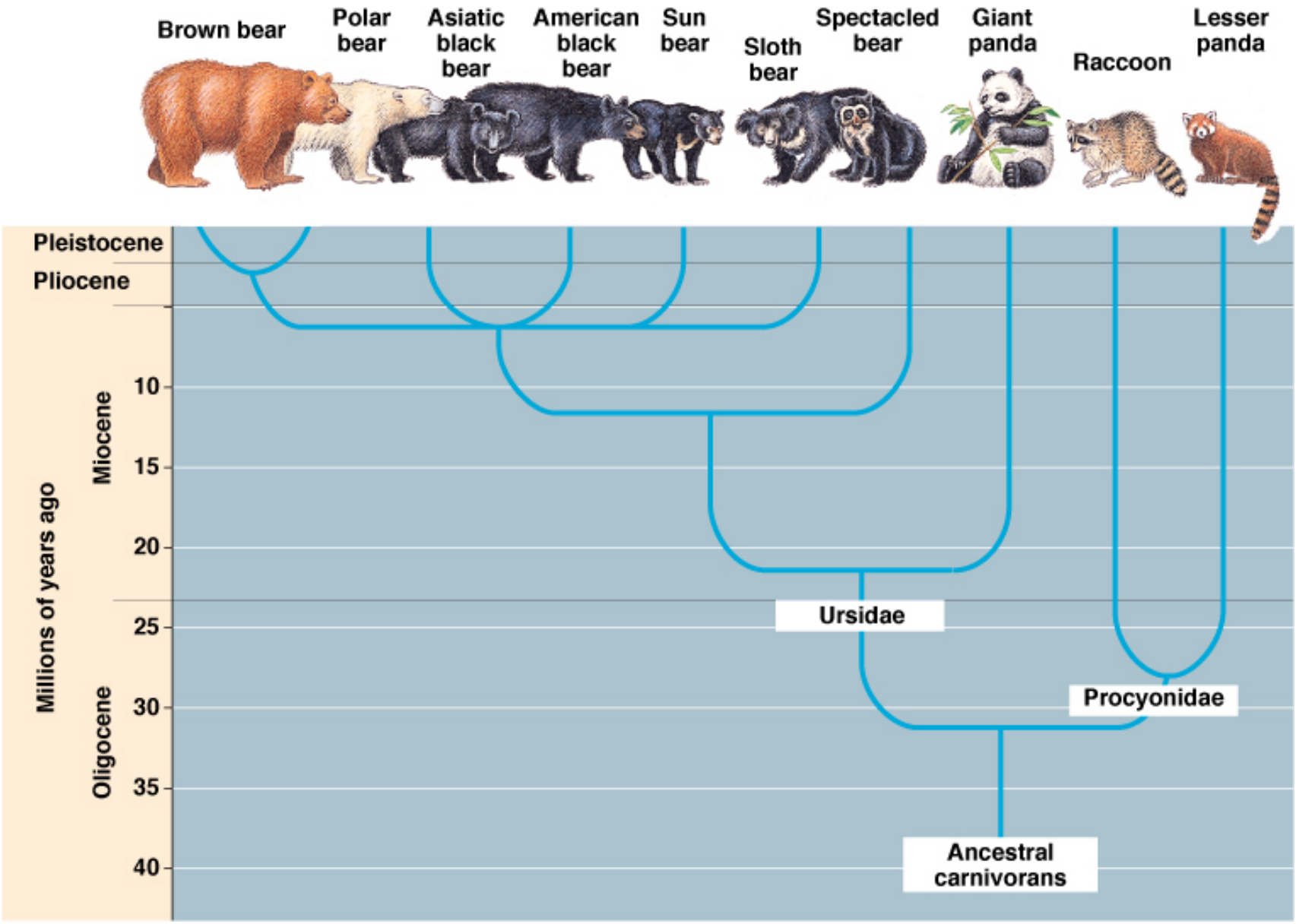
1. **Importance of Phylogeny**
2. Distance Based Trees



Giant Panda

- Is a giant panda a bear or a raccoon?
- When they were first discovered in Europe scientists couldn't decide—they had features of both
- They were named after red pandas that are raccoons
- This was only resolved in 1985 when Steven O'Brien and colleagues used sequence matching from 500 000 nucleotides to finally produce a definitive evolutionary tree

The Tree of Bears

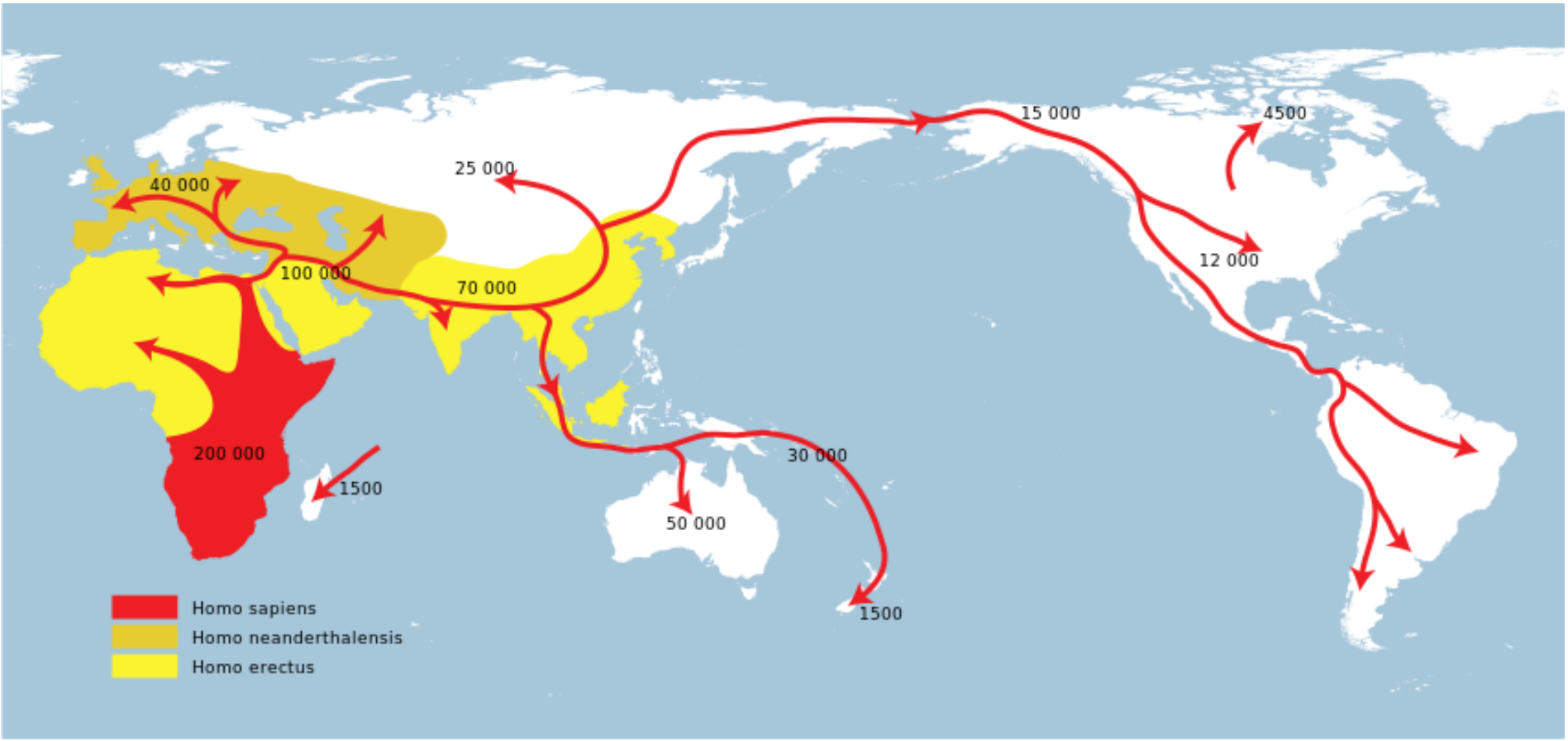


Out of Africa

- In the mid 1980s Allan Wilson and colleagues studied the evolutionary tree of human mitrochondria and decided that all races outside Africa shared a common ancestor who lived in Africa 200 000 years ago

- Hominids had left Africa much earlier (e.g. Neanderthals) so this theory postulated that modern humans would have displaced all existing hominids outside of Africa

Modern Humans



By NordNordWest - File:Spreading homo sapiens ru.svg by Urutseg, Public Domain, https://commons.wikimedia.org/w/index.php?curid=34697001

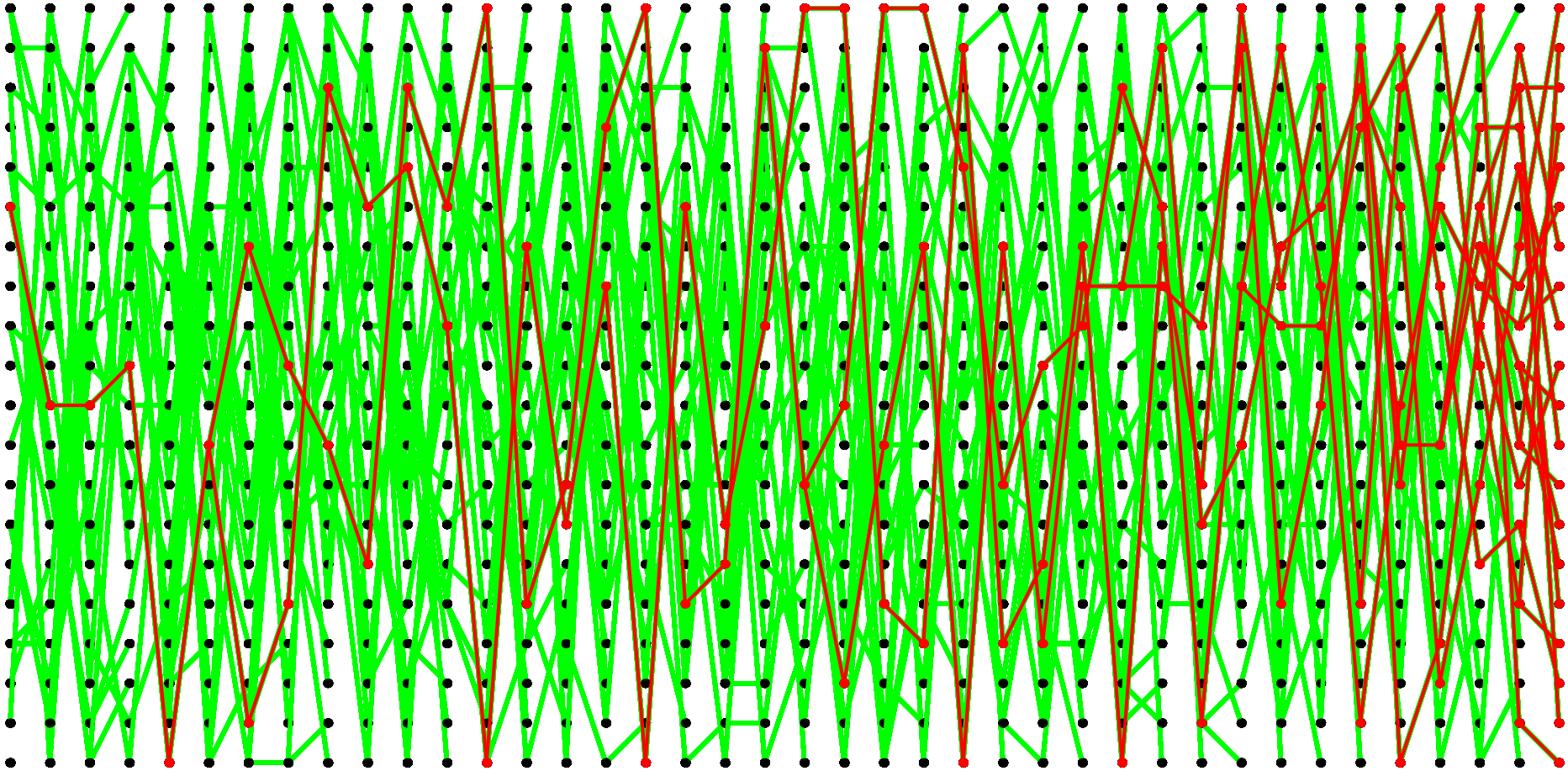
Mitrochondrial DNA

- Mitrochondria are the powerhouse of cells
- They would have originally been an independent organism that somehow got captured by eukaryotic cells
- They contain their own DNA
- They are inherited through the mother's ova
- They therefore reproduce asexually

Asexual Inheritance

- Asexual inheritance is much easier to study
- Populations will share a common ancestor because of
  - ⋆ Natural coalescence
  - ⋆ Going through a bottle neck (e.g. leaving Africa)
- Natural coalescence happens by chance, but will be enhanced by selection

Coalescence



- If you go far enough back in time an asexual population will have a common ancestor

Cross-Breeding

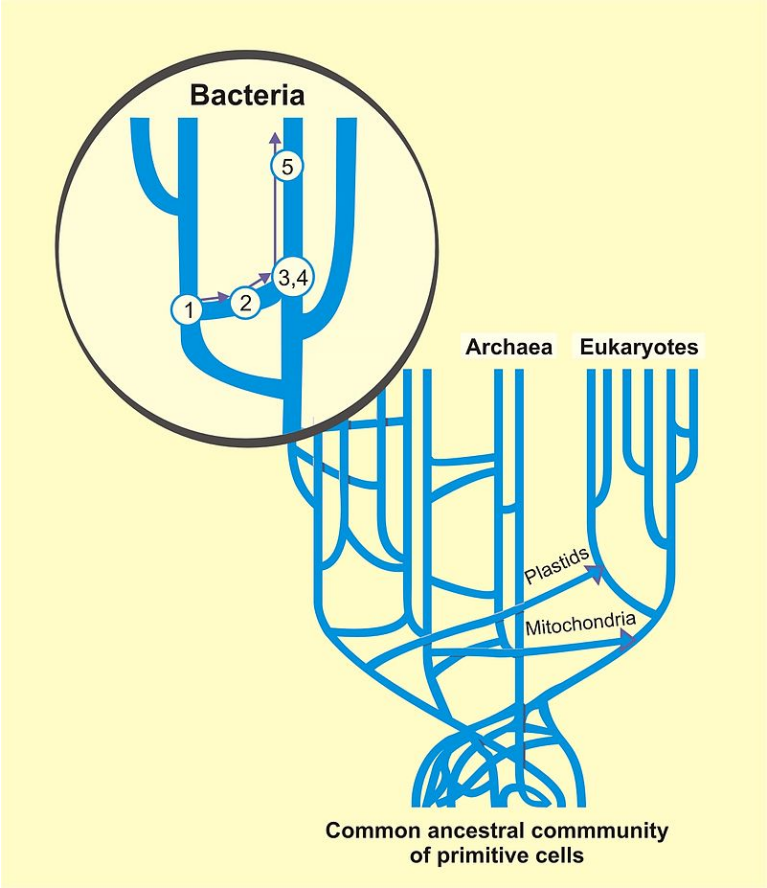
- Didn't modern man cross-breed with other hominids?
- This was unknown even 10 years ago, but of course they did
- However, the numbers were not large and due to coalescence (the coalescence time being small because there was not huge cross-breeding) all traces of Neanderthal mitrochondrial DNA vanished
- The story is very different for the rest of our DNA
- Normal DNA evolves sexually

The Selfish Gene

- With crossover you cannot build a tree of life as you inherit from two parents
- However, each gene (at least if it doesn't get disrupted by crossover) can be mapped to its own ancestral tree
- Because there are lots of genes it is much more likely for genes from different origins to survive (particularly if they have different fitness)
- In fact, we carry DNA that clearly came from Neanderthal and a few other early hominids
- These genes seem to have been selected for

Horizontal Gene Transfer

- Although bacteria and archaea reproduce asexually they regularly exchange plasmids
- These are loops of DNA
- This is the primary cause of antibiotic resistance
- The tree of life is not quite a tree



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Ligers and Tigons

- Most higher animals form species where individuals between species don't inter-breed very much
- In captivity ligers are crosses between male lions and female tigers and tigons are crosses between male tigers and female lions
- In the wild lions and tigers no longer overlap, but in the past there was some report of ligers (which are typically enormous)
- Like most cross-breeds they have reduced fertility (one sex is often inviable or sterile) so that cross breeds die out
- Because of this lack of significant cross-breeding we can model most of evolution at a species level as a tree

Evolution and HIV

- We often think of evolution as happening over geological time
- But the HIV virus evolved over decades from a disease that infected African primates to one that infects man
- Furthermore it evolves over months and years to evade our immune system
- Tracking these changes can be important in understanding and controlling the disease
- The virus evolves asexually so we can build an evolution tree

Evolution of Cancer

- Cells even in higher organisms predominantly reproduce asexually (they continuously divide)
- Over time the genome can get disrupted which if we are unlucky can grow into tumours
- Cancer cells accumulate mutations, some of which will die out others of which will replicate faster that other cells
- Understanding the development of cancer is often crucial to its treatment

Orthologues and Paralogues

- Genes that diverge due to speciation are known as orthologues
- Genes that diverge due to gene duplication are known as paralogues
- If we are interested in understanding the relationship between species (e.g. pandas) we have to make sure we choose orthologues
- But if we are interested in the different types of haemoglobins then we have to look at paralogues

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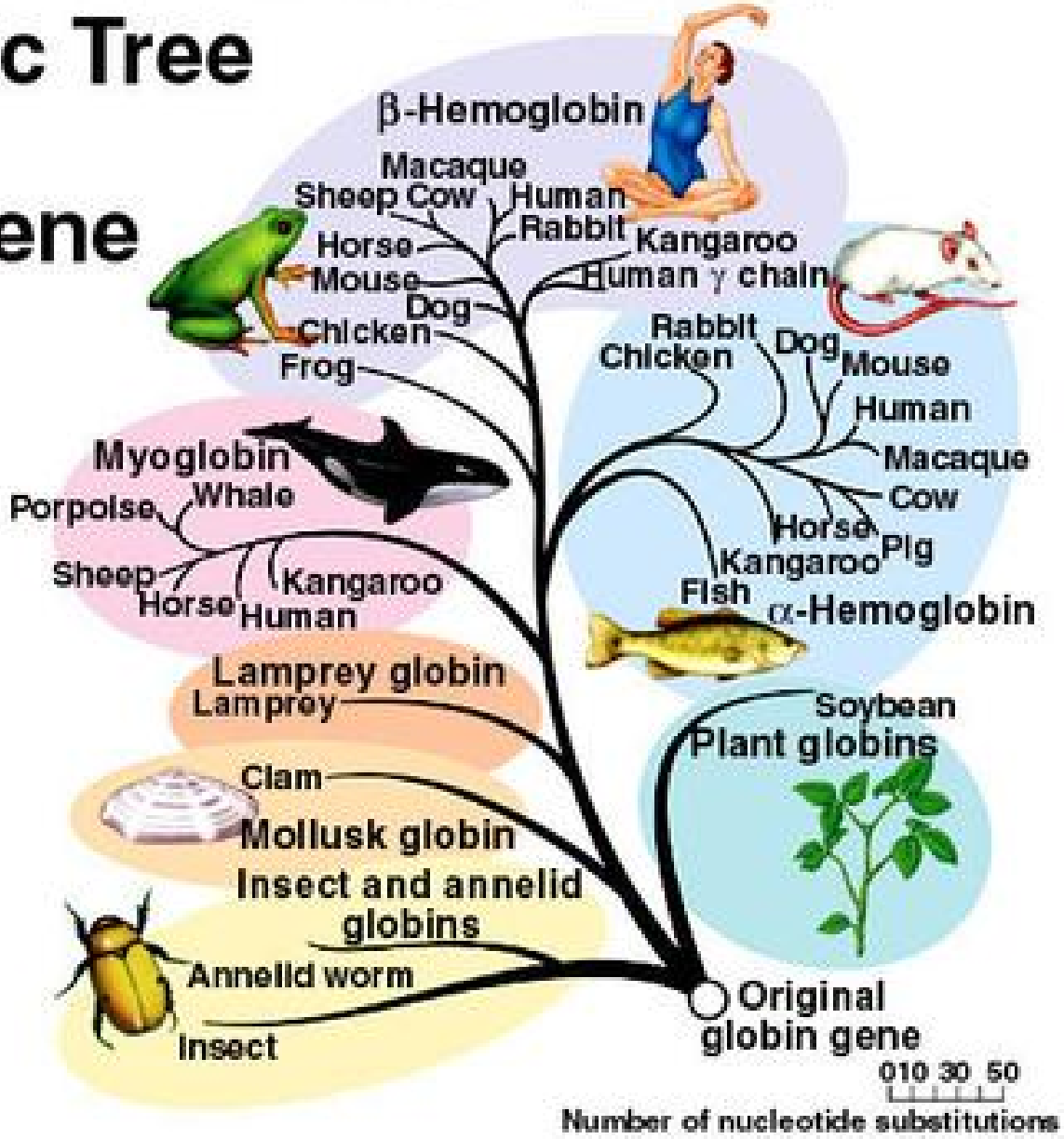
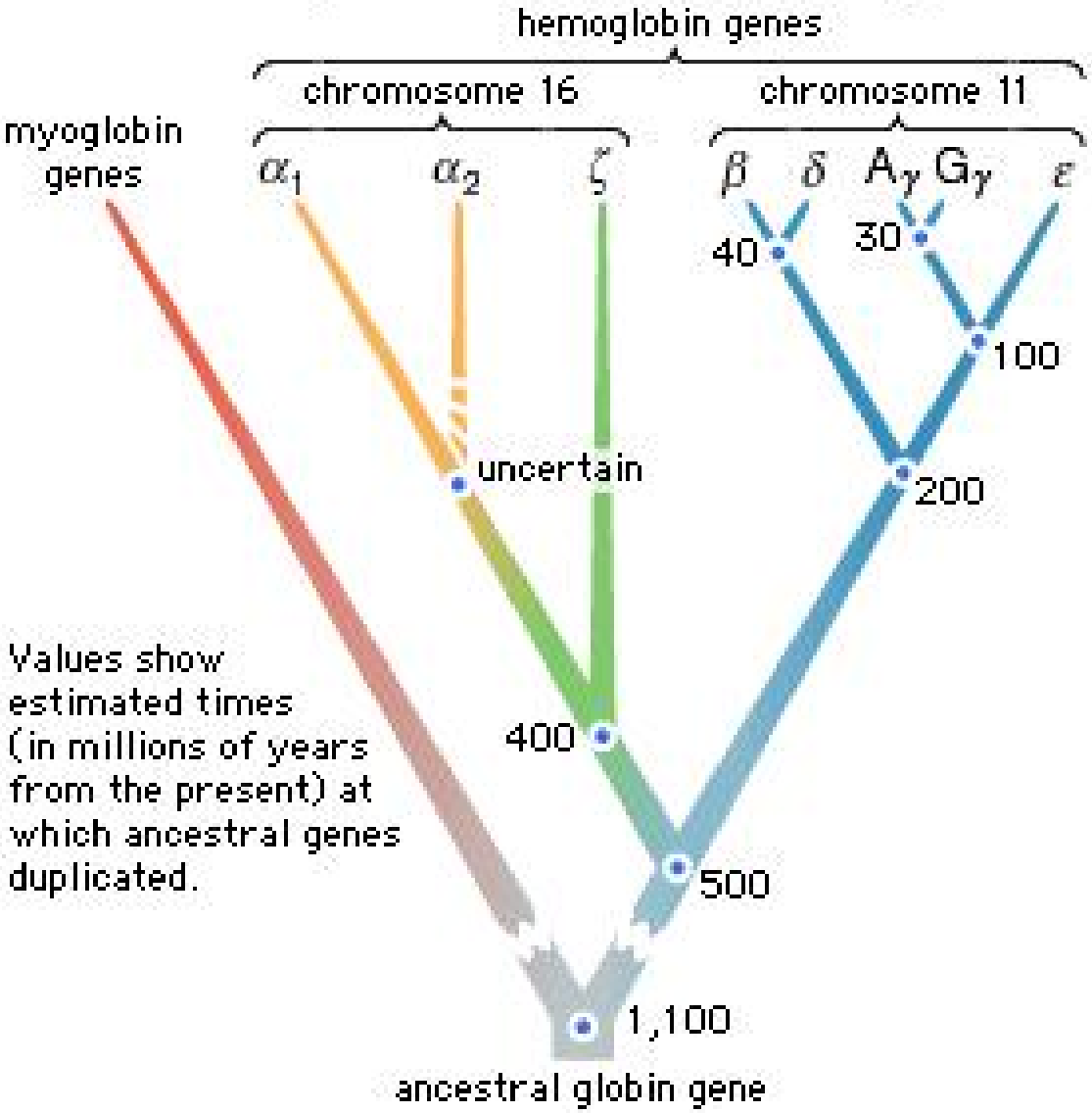
Phylogenetic Tree of Globin Gene



Evolutionary history of the globin genes



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Three Camps

- Although you would think that evolutionary trees should be pretty well understood, there is a huge debate about how to do this right
- There are three bitter rival camps in the field
  - ⋆ Evolutionary Taxonomy
  - ⋆ Phenetics or numerical taxonomy
  - ⋆ Cladistics
- If you are interested you can delve in the philosophical differences, but we will ignore them

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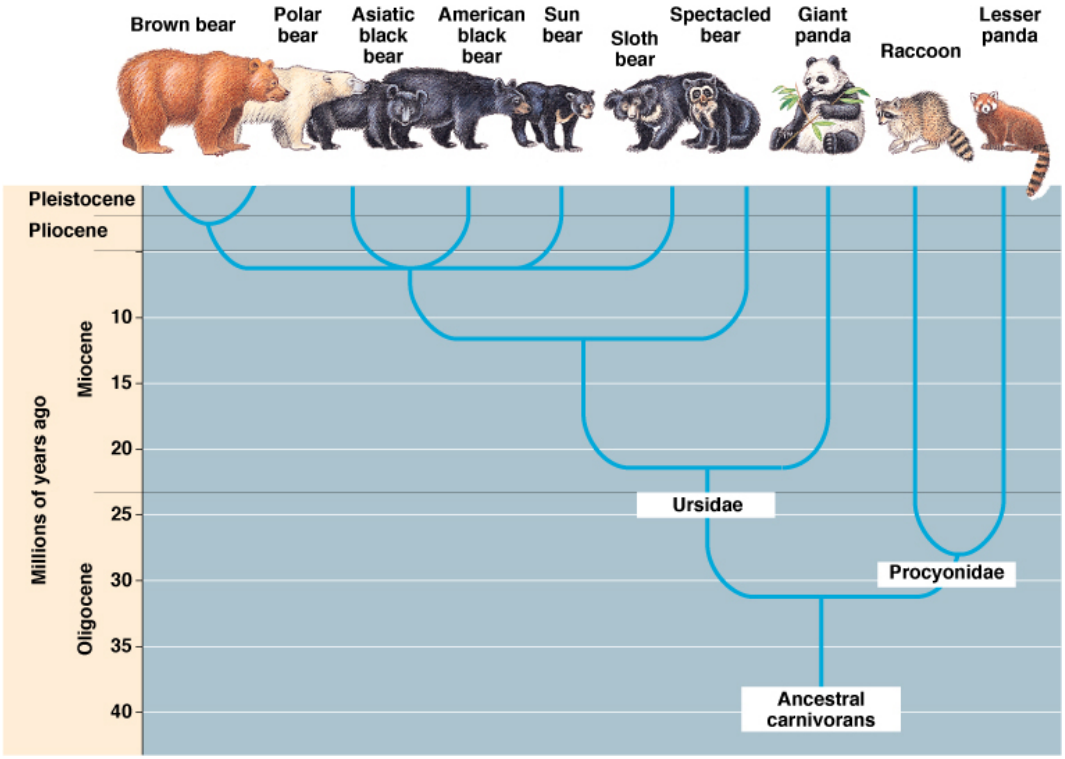


Building Phylogenetic Trees

- Building evolutionary trees (also known as phylogenetic trees) based on some measure of evolutionary distance is scientifically important
- Traditionally morphological features where used, but today sequence edit distance is usually used
- There are often many plausible tree that can explain the data
- Efficient methods that find the best evolutionary trees still don't exist—current algorithms are compromises

The Molecular Clock

- It is a nice picture to see the accumulation of mutations acts as a clock telling us when species separated
- Unfortunately, mutations are random and strongly affected by selection pressure
- Also mutations are often very complex, involve a duplication of a stretch of DNA
- A lot of care is needed to use mutations as a molecular clock, including using as large sequences as possible

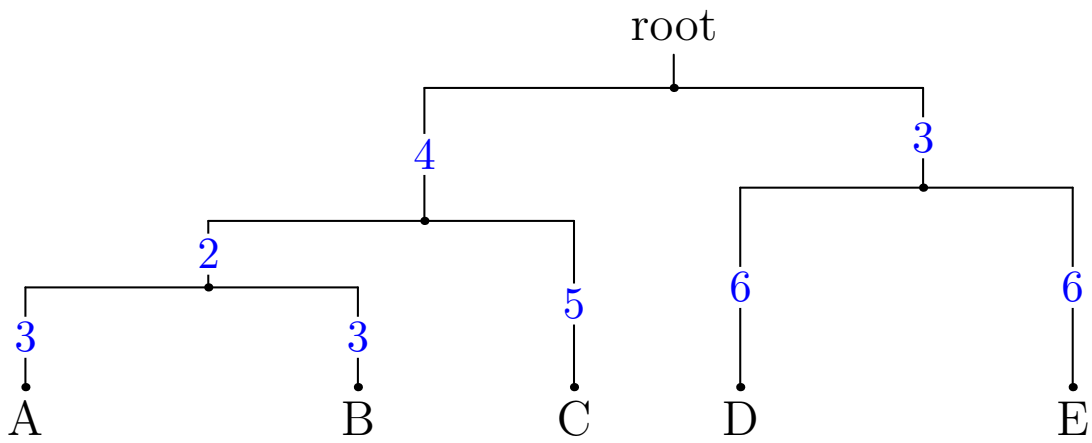
Two Types of Trees

- There are two main types of evolutionary trees
  - ⋆ **Distance-based trees**, where we have a table of "distances" between species and we wish to find a tree explaining these distances
  - ⋆ **Sequence based trees**, where we are given a set of sequences and we want to find a tree with a minimum number of mutations per link (maximum parsimony) that explains the data

Ultrametric Trees

- We consider binary trees

- If we had a perfect molecular clock then evolutionary trees to existing species will have an ultrametric structure



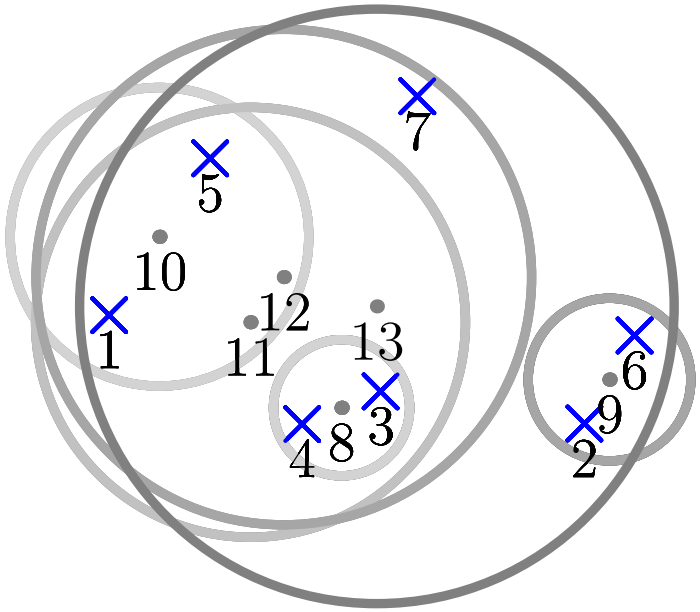
	A	B	C	D	E
A	0	3	5	9	9
B	3	0	5	9	9
C	5	5	0	9	9
D	9	9	9	0	6
E	9	9	9	6	0

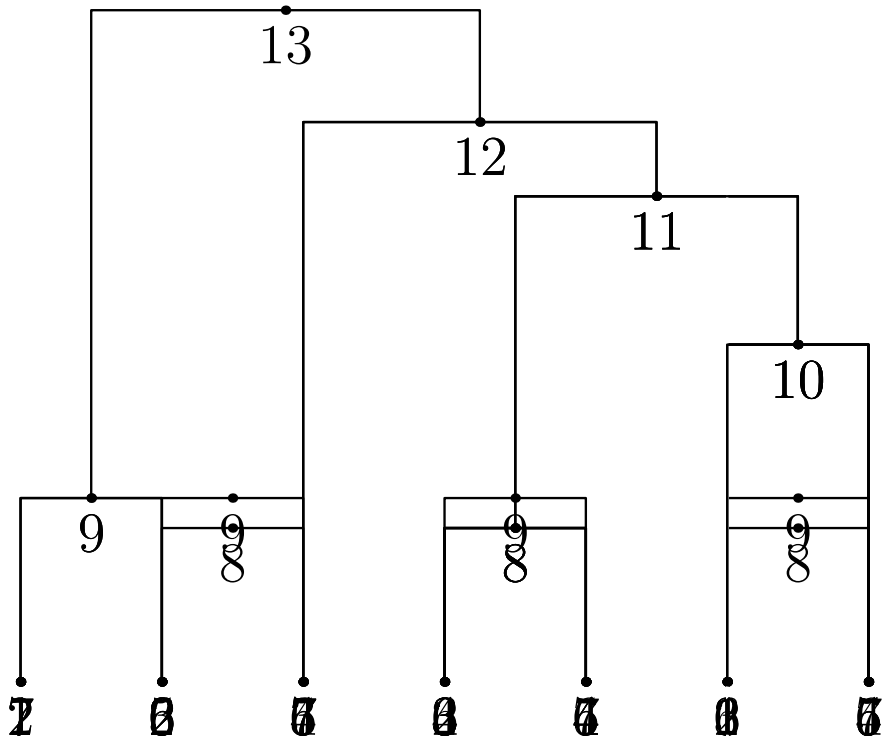
- For any three nodes i, j and k with distances d_{ij}, d_{ik} and d_{jk} two distances will be identical, and one will be smaller than these

Real Trees

- Because we cannot accurately measure divergence time, in practice it is very unlikely that the table of distances can be modelled with an ultrametric tree
- We can however build an ultrametric tree through a clustering algorithm known as **Unweighted Pair Group Method using Arithmetic averaging** UPGMA
- This is an easy algorithm where we build a tree from the leaves upwards
- We select the two closest subtrees remove all the nodes and replace the two joined subtree with a new subtree

UPGMA





Distance Calculation for UPGMA

- The UPGMA algorithm starts from a table of distances

- The distance between two clusters $\mathcal{C}_k$ and $\mathcal{C}_l$ is given by

$$d_{kl} = \frac{1}{|\mathcal{C}_k|\,|\mathcal{C}_l|} \sum_{\substack{p \in \mathcal{C}_k \\ q \in \mathcal{C}_l}} d_{pq}$$

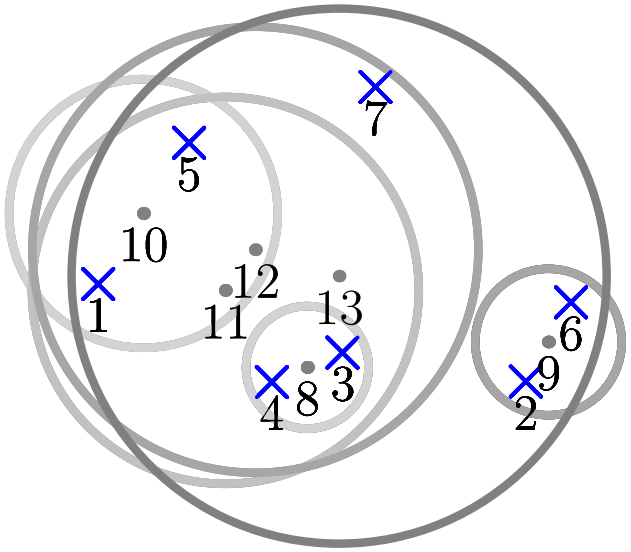
- If we join two clusters so that $\mathcal{C}_k = \mathcal{C}_i \cup \mathcal{C}_j$ then the distance to a cluster $\mathcal{C}_l$ is

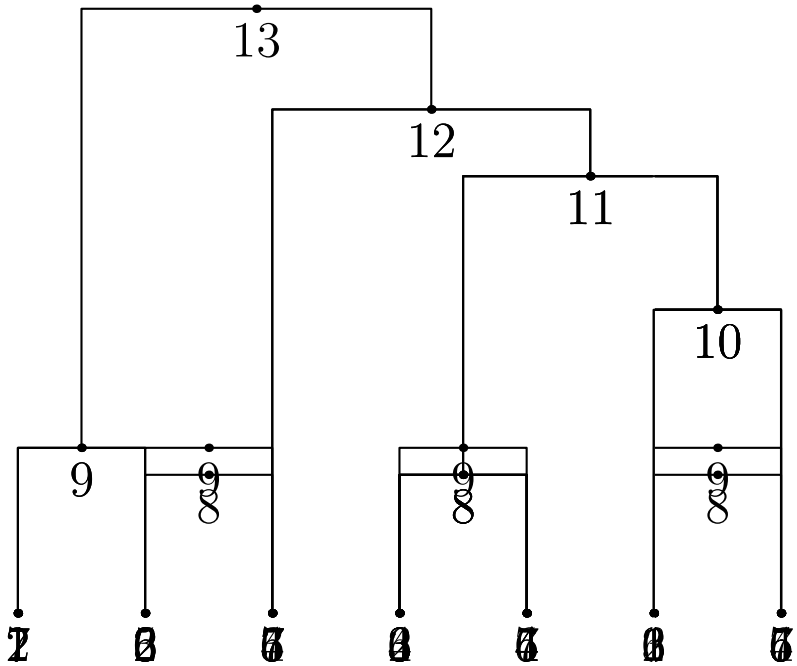
$$d_{kl} = \frac{d_{il}\,|\mathcal{C}_i| + d_{jl}\,|\mathcal{C}_j|}{|\mathcal{C}_i| + |\mathcal{C}_j|}$$

- If the distances are not ultrametric then UPGMA can give poor trees

Weakness of UPGMA

- Note that organism 5 and 7 are quite close, but UPGMA distorts this





Reflection on Ultrametricity

- UPGMA imposes an ultrametric structure on the derived evolutionary trees
- If mutations accumulated at a constant rate then evolutionary trees should have an ultrametric structure
- Unfortunately mutations experience selection and in some cases they are very non-trivial (genome copy mutations)
- There are stretches of "junk" DNA where mutations appear to accumulate at a reasonable constant rate, but these are not well studied or stable regions

Summary

- Evolutionary trees provide very important scientific insights
- Building evolutionary trees is non-trivial
- If the distances are approximately ultrametric then UPGMA will generate a good tree
- Need to look at other ways of building evolutionary trees